

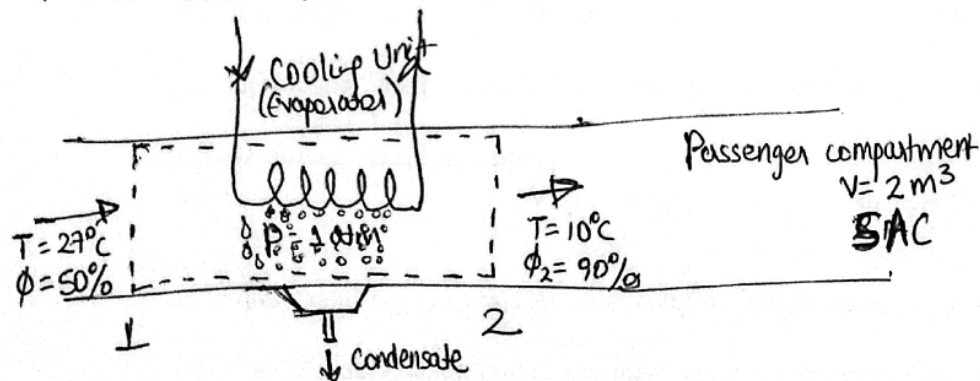
2. Gas vapor mixture

Example 2-1

Atmospheric air from the inside of an automobile enters the evaporator section of the air conditioner at 1 atm, 27°C , & 50 percent relative humidity. The air returns to the automobile at 10°C & 90 percent relative humidity. The passenger compartment has a volume of 2 m^3 and five air changes per minute are required to maintain the inside of the automobile at the desired comfort level. Sketch the psychrometric diagram for the atmospheric air flowing through the air-conditioning process. Determine the dew-point & wet-bulb temperatures at the inlet to the evaporator section, in $^\circ\text{C}$. Determine the required heat transfer rate from the atmospheric air to the evaporator fluid, in kW. Determine the rate of condensation of water vapor in the evaporator section, in kg/min .

GivenSolution

Automobile Air conditioner

Required

- Sketch the psychrometric diagram
- T_{dp} , T_{wb} at inlet
- $T_{dp,1}$, $T_{wb,1}$ $^\circ\text{C}$
- $\dot{Q}_{out}$
- $\dot{m}_w$

Assumptions

- Steady flow process
 - ↳ mass flow rate of dry air remains constant
 - $\dot{m}_{a,1} = \dot{m}_{a,2} = \dot{m}_a$
- Dry air & water vapor are ideal gases.
- ΔKE & ΔPE are negligible, $\cong 0$

Analysis

page 2-2 //

Energy balance on control volume: $\frac{dE}{dt} = \sum \dot{E}_{in} - \sum \dot{E}_{out}$

$$\sum \dot{E}_{in} = \sum \dot{E}_{out}$$

$$\dot{m}_a = \dot{m}_{a2} = \dot{m}_a \text{ --- steady flow}$$

$$\dot{m}_a h_1 = \dot{Q}_{out} + \dot{m}_a h_2 + \dot{m}_w h_{w2} \text{ --- (1)}$$

Where from $V = \frac{V}{m}$

$$\dot{m}_a = \frac{\dot{V}_1}{V_1} = \frac{V_{can} ACH}{V_1}$$

N.B ACH - Air changes per hour

* The number of times that the total air volume in a space is completely removed and replaced in an hour.

where V_1 from psychrometric chart:

$$T_{db} = 27^\circ C$$

$$\phi = 50\%$$

$$T_{dp1} = 15.7^\circ C$$

$$T_{wb1} = 19.5^\circ C$$

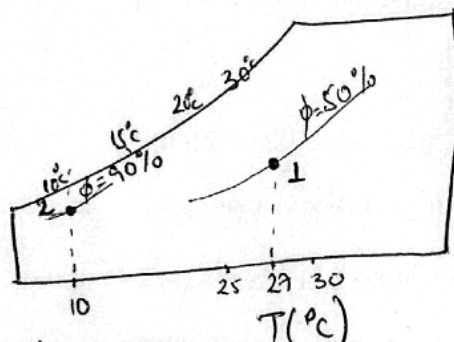
$$h_1 = 55.60 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.01115 \text{ kg H}_2\text{O/kg dry air}$$

$$V_1 = 0.8655 \text{ m}^3/\text{kg dry air}$$

$$h_2 = 27.35 \text{ kJ/kg dry air}$$

$$\omega_2 = 0.00686 \text{ kg H}_2\text{O/kg dry air}$$



Unable to connect pt 1 & 2, since the process is unknown.

$$\Rightarrow \dot{m}_a = \frac{V_{can} ACH}{V_1} = \frac{(2 \text{ m}^3/\text{change}) * (5 \text{ changes/min})}{0.8655 \text{ m}^3/\text{kg dry air}}$$

$$= 11.55 \text{ kg/min}$$

$$\dot{m}_w \text{ } h_{w2} \text{ } ?$$

$$\text{Rate of condensation of water vapor } \dot{m}_w = \dot{m}_{v1} - \dot{m}_{v2}$$

$$= 0.1288 \text{ kg/min} - 0.07926 \text{ kg/min}$$

$$= 0.0495 \text{ kg/min}$$

Substituting in eqn (1)

$$\dot{m}_a h_1 = \dot{Q}_{out} + \dot{m}_a h_2 + \dot{m}_w h_{w2}$$

$$\dot{Q}_{out} = 324.4 \text{ kJ/min} = 324.4 \text{ kJ/min} * \frac{1 \text{ min}}{60 \text{ sec}} = 5.41 \text{ kW}$$

Mass flow rate of vapor: $\omega = \frac{\dot{m}_w}{\dot{m}_a}$

$$\dot{m}_{v1} = \omega_1 \dot{m}_a = (0.01115) * (11.55 \text{ kg/min})$$

$$\dot{m}_{v2} = \omega_2 \dot{m}_a = 0.1288 \text{ kg/min}$$

$$= (0.00686 \text{ kg/kg dry air}) * (11.55 \text{ kg/min})$$

$$= 0.07926 \text{ kg/min}$$

Example

1. A room that contains air @ 1 atm, 35°C, & 40% ϕ

Determine: ω

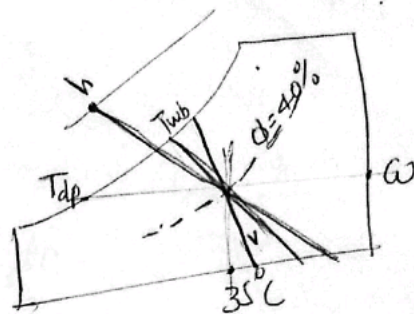
h

T_{wb}

T_{dp}

v

Chart prepared



page 2-2-1

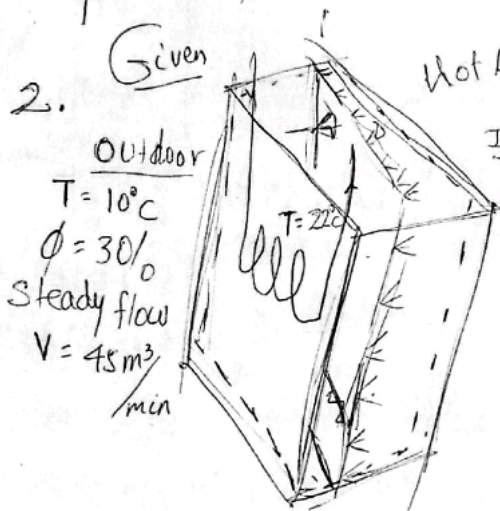
Analysis

2 givens & necessary

$T_{db} = 35^\circ\text{C}$

$\phi = 40\%$

Given



Hot Steam

Indoor

$T = 25^\circ\text{C}$
 $\phi = 60\%$

Example 2-2

An Air-conditioning system is to take in outdoor air at 10°C & 30 percent relative humidity at a steady rate of 45 m³/min & to condition it to 25°C & 60 percent relative humidity.

The outdoor air is first heated to 22°C in the heating section & then humidified by the injection of hot steam in the humidifying section.

$P = C = 100 \text{ kPa}$

Determine Required:

a. $\dot{Q}_{\text{heating section}}$

b. $\dot{m}_{\text{required in the humidifying section}}$

Assuming the entire process takes place at a pressure of 100 kPa, determine a. the rate of heat supply in the heating section &

b. the mass flow rate of the steam required in the humidifying section.

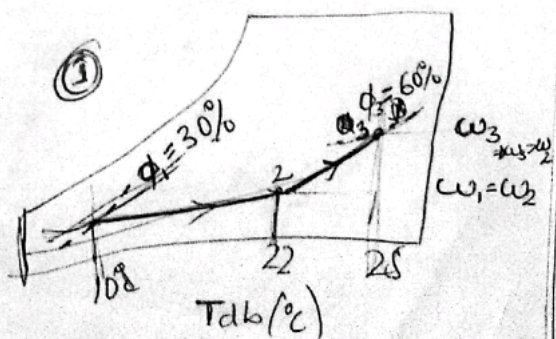
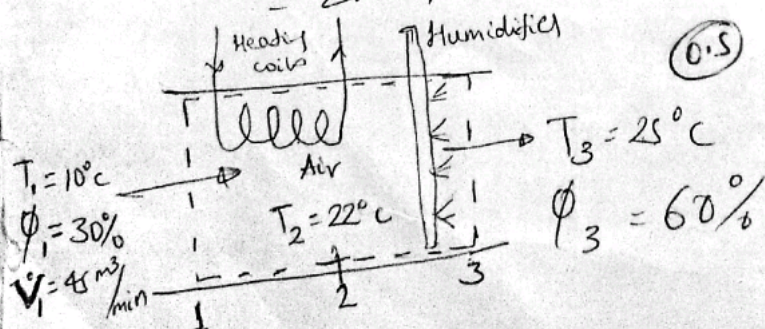
Analysis

Assumptions

- steady flow process.
- Mass flow rate of dry air is constant: $\dot{m}_a = \dot{m}_a = \dot{m}_a$
- Dry air & water vapor are ideal gases.
- $\Delta KE, \Delta PE \approx 0$ (are negligible)

(*) Psychrometric chart

* $P = 1 \text{ atm} = 101.325 \text{ kPa}$



$\omega_1 = \omega_2$

a. Qin

Energy balance: $\dot{Q}_{in} + \dot{m}_a(h_1 + \frac{V_1^2}{2} + gz_1) + \dot{W}_{in} = \dot{Q}_{out}$

$$\dot{Q}_{in} + \dot{m}_a h_1 = \dot{m}_a h_2$$

$$\dot{Q}_{in} = \dot{m}_a (h_2 - h_1)$$

$$= 67.3 \text{ kJ/min}$$

(0.25)

mass balance for dry air:

$$\dot{m}_a = \dot{m}_2 = \dot{m}_a$$

From $h = h_a + \omega h_v$ for air

$$h_1 = c_p T_1 + \omega_1 h_{g1}$$

$$= 15.8 \text{ kJ/kg dry air}$$

$$h_2 = c_p T_2 + \omega_2 h_{g2}$$

$$= 28.0 \text{ kJ/kg dry air}$$

(0.25)

$$\omega_1 = 0.622 \frac{P_{v1}}{P_1 - P_{v1}}$$

$$= 0.00231 \text{ kg H}_2\text{O/kg dry air}$$

$$\omega_1 = \omega_2$$

(0.25)

$$\left(v = \frac{V}{m} \right) (P_v = R_a T)$$

$$\left| \frac{\dot{m}_a}{V_1} \right| \left| \frac{V_1}{P_a} \right| \left| \frac{P_a}{P_1} \right| \left| \frac{P_1}{P_1 - P_{v1}} \right|$$

$$= \frac{0.289 \times 283}{99.63} = 0.815 \text{ m}^3/\text{kg dry air}$$

$$= 55.2 \text{ kg} = 0.815 \text{ m}^3/\text{kg dry air}$$

$$= 0.00231 \text{ kg H}_2\text{O/kg dry air}$$

$$\omega_1 = \omega_2$$

$$= 0.00231 \text{ kg H}_2\text{O/kg dry air}$$

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data from

At 100m temp

$$\rho = 1.005 \text{ kg/m}^3$$

$$R_a = 0.287 \text{ kJ/kg K}$$

$$P_{sat}@10^\circ\text{C} = 1.2281 \text{ kPa}$$

$$P_{sat}@25^\circ\text{C} = 3.1698 \text{ kPa}$$

$$h_{sat, \text{water vapor}}$$

$$@10^\circ\text{C} = 2519.2 \text{ kJ/kg}$$

$$h_{sat}@22^\circ\text{C} = 2541.0 \text{ kJ/kg}$$

$$V_1 = 0.815 \text{ m}^3/\text{kg dry air}$$

$$h_1 = 15.8 \text{ kJ/kg}$$

$$h_2 = \text{no two data}$$

$$27.8 \text{ or } 27.9$$

$$h_2 \text{ from } \omega_2 \text{ \& } T_2$$

$$2 \text{ marks for psychrometric chart}$$

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mass to

Humidify section mass balance:

$$\dot{m}_a \omega_2 + \dot{m}_w = \dot{m}_a \omega_3$$

$$\dot{m}_w = \dot{m}_a (\omega_3 - \omega_2)$$

$$\text{where } \omega_3 = 0.622 \frac{\phi_3 P_{g3}}{P_3 - \phi_3 P_{g3}}$$

$$= 0.01206 \text{ kg H}_2\text{O/kg dry air}$$

$$\dot{m}_w = \dot{m}$$

$$\dot{m}_w = 0.539 \text{ kg/min}$$

Discussion

$$1 \text{ day} = 0.539 \text{ kg/min} \times 60 \text{ min/hr} \times 24 \text{ hr/day}$$

$$= 776.16 \text{ kg/day} \approx 1 \text{ ton/day} \Rightarrow \text{which is significant}$$

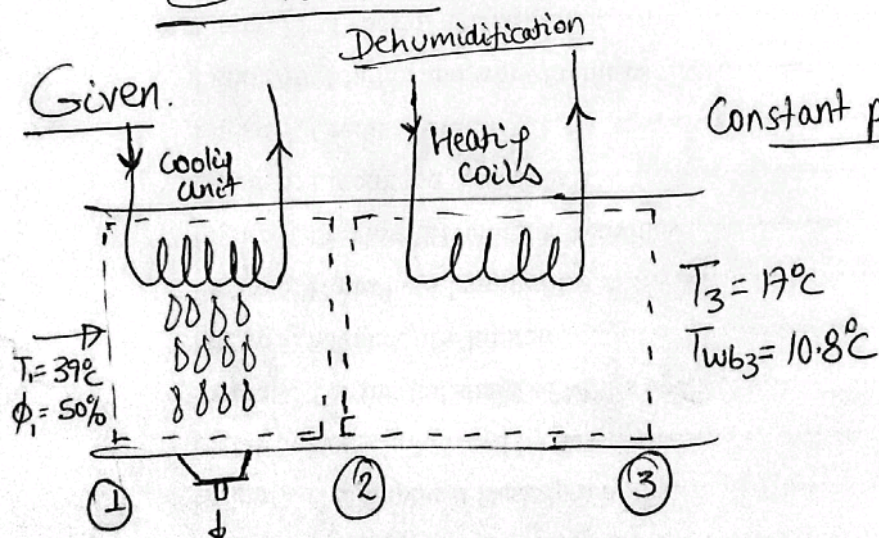
(0.25)

Example 2-3

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Humid air ^{in New York} is to be conditioned in a constant-pressure process at 1 atm from 39°C dry bulb & 50 percent relative humidity to 17°C dry bulb & 10.8°C wet bulb. The air is first passed over cooling coils to remove all of the moisture necessary to achieve the final moisture content & then is passed over heating coils to achieve the final state.

- Sketch the psychrometric diagram for the process.
- Determine the dew-point temperature of the mixture at the inlet of the cooling coils & at the inlet of the heating coils.
- Determine the heat removal by the cooling coils, the heat addition by the heating coils, and the net heat transfer for the entire process, all in kJ/kg dry air .

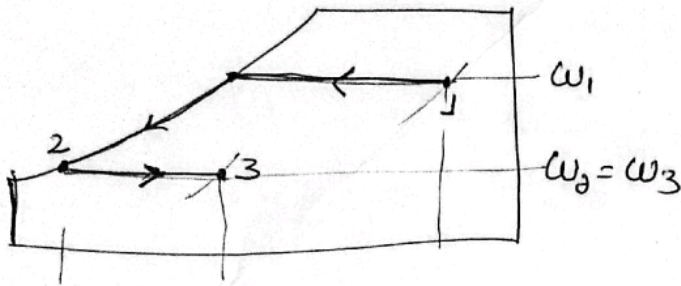
SolutionRequired

- Sketch the process
- T_{dp1}
 T_{dp2}
- Q_{out}

Assumptions

- Steady flow process
→ $\dot{m}_a$ is constant
- Dry air & water vapor are ideal gases
- $\Delta KE, \Delta PE \approx 0$

a.

b. $p = 1 \text{ atm}$, use psychrometric chart

At ① $T_1 = 39^\circ\text{C}$
 $\phi = 50\%$
 $p = 1 \text{ atm} = 101.325 \text{ kPa}$

$T_{dp1} = 26.7^\circ\text{C} \checkmark$
 $h_1 = 96.43 \text{ kJ/kg dry air}$
 $\omega_1 = 0.02225 \text{ kg H}_2\text{O/kg dry air}$

At ③ $T_3 = 17^\circ\text{C}$
 $T_{wb3} = 10.8^\circ\text{C}$
 $p = 1 \text{ atm} = 101.325 \text{ kPa}$

$T_{dp3} = 5.3^\circ\text{C}$
 $\phi_3 = 46.0\%$
 $h_3 = 31.03 \text{ kJ/kg of dry air}$
 $\omega_3 = 0.005514 \text{ kg H}_2\text{O/kg dry air}$

 $\Rightarrow$ Cooling

$\omega_2 = \omega_3 = 0.005514 \text{ kg H}_2\text{O/kg of dry air}$

$\phi_2 = 1 = 100\%$

$p = 101.325 \text{ kPa}$

$T_{dp2} = 5.3^\circ\text{C} \checkmark$
 $T_2 = 5.3^\circ\text{C}$
 $h_2 = 19.16 \text{ kJ/kg dry air}$

$\Rightarrow h_w \approx h_f @ 5.3^\circ\text{C} = 22.25 \text{ kJ/kg}$ — Table A-4

↑ temp. of air at the end of dehumidification process.

c. Cooling and dehumidification process

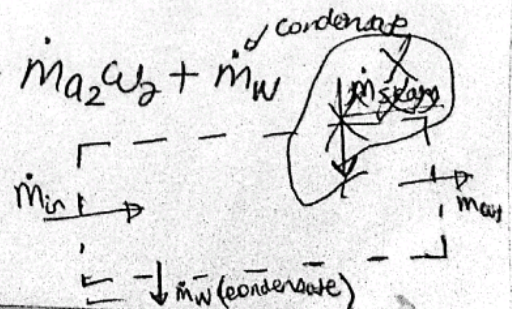
Mass Balance for Water

$\sum \dot{m}_{w,i} = \sum \dot{m}_{w,e} \Rightarrow \dot{m}_a \omega_1 = \dot{m}_a \omega_2 + \dot{m}_w$

$\dot{m}_w = \dot{m}_a (\omega_1 - \omega_2)$

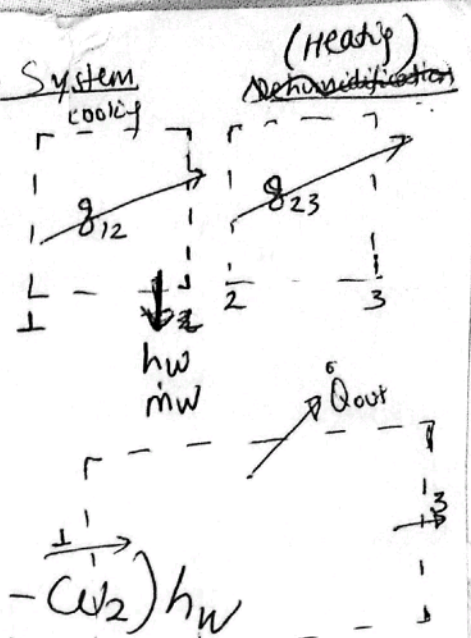
Energy balance:

$\dot{E}_{in} - \dot{E}_{out} = \Delta \dot{E}_{system} = 0$ (steady)



$$\dot{E}_{in} = \dot{E}_{out}$$

#page 2-2-33



$$\sum \dot{m}_i h_i = \dot{Q}_{out} + \sum \dot{m}_e h_e$$

$$\dot{Q}_{out} = \dot{m}_a h_1 - (\dot{m}_a h_2 + \dot{m}_w h_w)$$

$$= \dot{m}_a (h_1 - h_2) - \dot{m}_w h_w$$

$$= \dot{m}_a (h_1 - h_2) - \dot{m}_a (w_1 - w_2) h_w$$

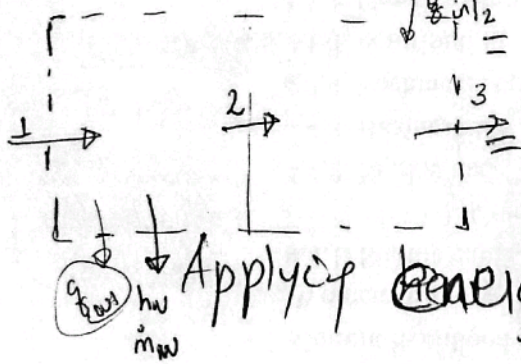
Substituting: $\frac{\dot{Q}_{12}}{\dot{m}_a}$

Heat removed in cooling section
 $\dot{E}_{in} - \dot{E}_{out} = \dot{Q}_{cooling} = 0$
 $\dot{E}_{in} = \dot{E}_{out}$

$$g_{12} = (h_1 - h_2) = (w_1 - w_2) h_w$$

$$(96.43 - 19.16) \text{ kJ/kg} = (0.0225 - 0.00554) \times 22.27 \text{ kJ/kg}$$

$$76.9 \text{ kJ/kg}$$



Applying energy balance to heating section:

$$g_{23} = h_3 - h_2 = (31.03 - 19.16) \text{ kJ/kg}$$

$$11.9 \text{ kJ/kg}$$

$$\dot{Q}_{in} + \dot{m}_a h_2 = \dot{m}_a h_3$$

∴ Net heat transfer is since the process is dehumidification

$$g_{23} = g_{12} - g_{23}$$

$$= (76.9 - 11.9) \text{ kJ/kg}$$

$$= 65.0 \text{ kJ/kg}$$

$$W_{net} = W_{out} - W_{in}$$

$$Q_{net} = Q_{in} - Q_{out}$$

$$W_{in} + Q_{in} = W_{out} + Q_{out}$$

$$Q_{in} - Q_{out} = W_{out} - W_{in}$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{Q}_{in} + \dot{m}_a h_2 = \dot{m}_a h_3$$

$$g_{in} = h_3 - h_2$$

Example 2-2

An Air-conditioning system is to take in outdoor air at 10°C & 30%^(percent) relative humidity at a steady rate of $45\text{ m}^3/\text{min}$ & to condition it to 25°C & 60 percent relative humidity.

The outdoor air is first heated to 22°C in the heating section & then humidified by the injection of hot steam in the humidifying section.

Assume the entire process takes place at a pressure of 100 kPa , determine:

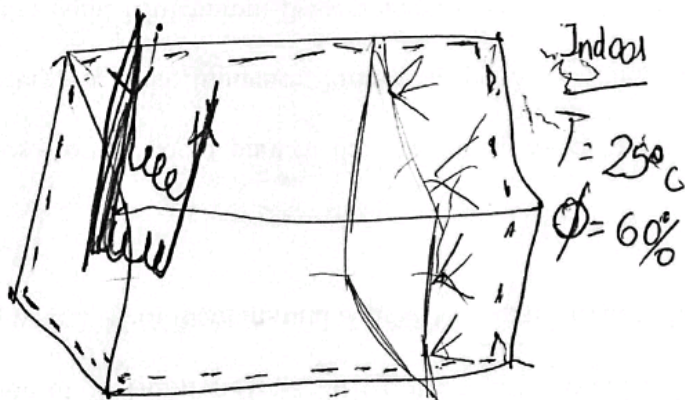
- the rate of heat supply in the heating section &
- the mass flow rate of the steam required in the humidifying section

Solution

Given

Outdoor

$T = 10^{\circ}\text{C}$
 $\phi = 30\%$
Steady flow
 $V = 45\text{ m}^3/\text{min}$



Life is experience
 Experience is everything
 May be the most sleep one is the most intelligent
 - Kunal

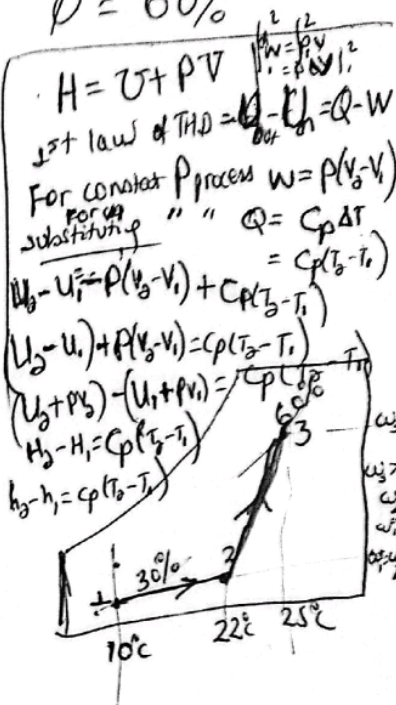
Solution

1st law of TMD: $\Delta U = Q - W$
 Change in heat internal energy added to the system

Indoor

$$T = 25^\circ\text{C}$$

$$\phi = 60\%$$



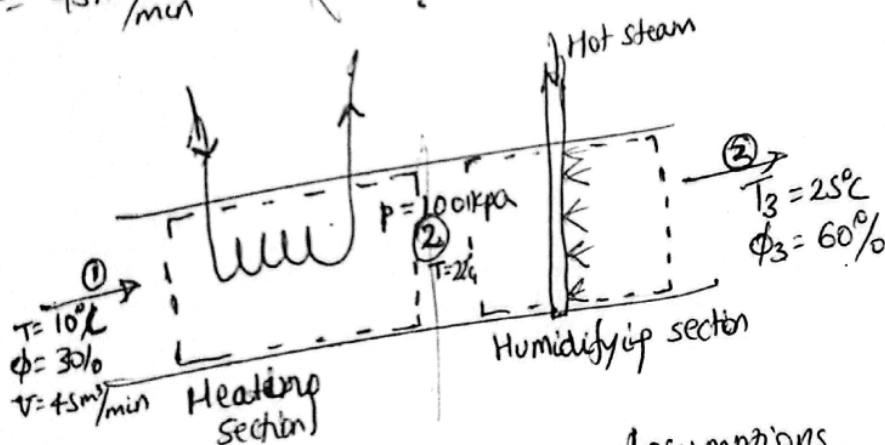
Outdoor

$$T = 10^\circ\text{C}$$

$$\phi = 30\%$$

Steady flow

$$\dot{V} = 45 \text{ m}^3/\text{min}$$



Required

a. $\dot{Q}_{in}$ (heating section)

b. $\dot{m}$ humidifier

Analysis

a. $\dot{Q}_{in}$ (heating section)

Energy balance

for control volume:

$$\frac{dE_{system}}{dt} = \sum \dot{E}_{in} - \sum \dot{E}_{out}$$

$$\sum \dot{E}_{in} = \sum \dot{E}_{out}$$

$$\dot{Q}_{in} + \dot{m}_1(h_1 + \frac{V_1^2}{2} + gz_1) = \dot{m}_2(h_2 + \frac{V_2^2}{2} + gz_2)$$

$$\dot{Q}_{in} + \dot{m}_1 h_1 = \dot{m}_2 h_2$$

$$\dot{Q}_{in} = \dot{m}_a(h_2 - h_1)$$

$$\dot{m}_1 = \dot{m}_a = \dot{m}_2$$

Assumptions

- Steady flow process
- Mass flow rate of dry air is constant
 $\dot{m}_{a1} = \dot{m}_{a2}$
- Dry air & water vapor are ideal gases.
- ΔKE & $\Delta PE \approx 0$ (negligible).

(*)

$$h(T) = \int_{T_{ref}}^T c_p(T) dT + h(T_{ref})$$

$$h = 0 \text{ at } T_{ref} = 0^\circ\text{C}$$

$$h_2 - h_1 = c_p(T_2 - T_1)$$

$$h_3 - h_1 = c_p(T_3 - T_1) + h_{fg}$$

$$h_3 - h_1 = c_p(T_3 - T_1)$$

$$h_1 = c_p(T_1 - T_{ref}) + h_{ref}$$

$$H = h_a + h_v = \dot{m}_a h_a + \dot{m}_v h_v$$

$$h = h_a + \omega h_v$$

Enthalpy for ideal gas independent of P & V depends only on its T .
 $\rightarrow$ Thermal energy

Real gases often closely approximate this behavior which simplifies practical thermodynamic design & analysis.

(*)